



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Advanced)

FULL SYLLABUS

29-03-2023

JEE(Main + Advanced) : ENTHUSIAST COURSE : [SCORE-II (ALL STAR BATCH)]

ANSWER KEY

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A,D	A,D	A,B,C,D	B,C	A,C	A,B
SECTION-I (ii)	Q.	7	8	9	10		
	A.	B	A	B	A		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	64.00	8.00	1.10 to 1.12	4.43 to 4.45	1.20 to 1.30	0.40
SECTION-II (ii)	Q.	7	8	9			
	A.	3	8	8			

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A,B,C	A,B,C	A,D	A,C,D	B,C,D	A,B,C
SECTION-I (ii)	Q.	7	8	9	10		
	A.	A	C	B	D		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	26.00	5.00	75.38	29.00	2.15	8.00
SECTION-II (ii)	Q.	7	8	9			
	A.	5	7	6			

PART-3 : MATHEMATICS

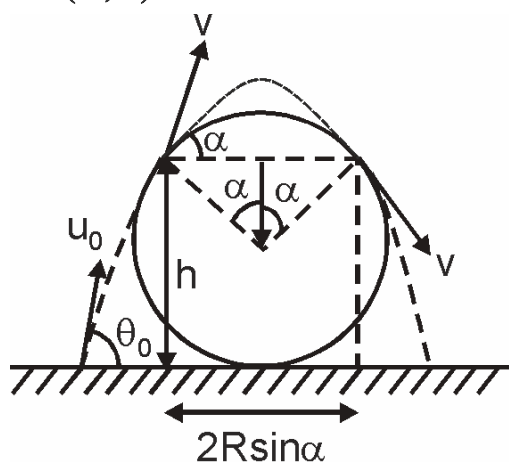
SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	B,D	A,B,D	A,D	A,B,C	A,B,C	A,B,D
SECTION-I (ii)	Q.	7	8	9	10		
	A.	A	B	B	A		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	5.00	18.00	1.50	2.00	300.00	0.45
SECTION-II (ii)	Q.	7	8	9			
	A.	4	2	5			

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (A,D)



For u to be minimum, it just grazes the cylinder at two points as shown

$$\text{From COME } \frac{mu^2}{2} = \frac{mv^2}{2} + mgh$$

$$u^2 = v^2 + 2gh$$

$$u^2 = v^2 + 2gR(1 + \cos\alpha)$$

$$\text{Also } 2R\sin\alpha = v^2 \frac{\sin 2\alpha}{g} \Rightarrow v^2 = \frac{Rg}{\cos\alpha} \quad \dots (2)$$

$$\text{From (1) and (2) } u^2 = Rg \left[2 + 2\cos\alpha + \frac{1}{\cos\alpha} \right]$$

$$\text{For } u \rightarrow \min, \frac{d}{d\alpha}(u^2) = 0$$

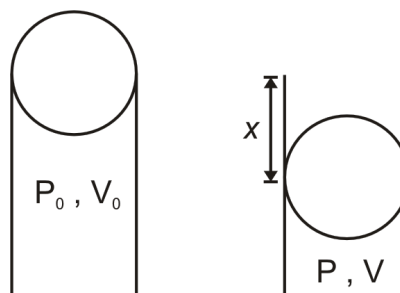
$$\Rightarrow \cos\alpha = \frac{1}{\sqrt{2}} \Rightarrow \alpha = 45^\circ \quad u_0^2 = Rg [2 + 2\sqrt{2}]$$

$$\text{Also } u\cos\theta = v\cos\alpha$$

$$\tan^2 \theta_0 = [3 + 2\sqrt{2}]$$

2. Ans (A,D)

(a) For isothermal compression of air



$$PV = P_0V_0$$

$$PA(L-x) = P_0AL \Rightarrow P = P_0 \left(\frac{L}{L-x} \right)$$

Work done by the gas on the ball

$$= -P_0V_0 \ln \left(\frac{L}{L-x} \right)$$

Work done by the gravity on the ball = mgx

If the ball comes to rest :

$$mgx - P_0V_0 \ln \left(\frac{L}{L-x} \right) = 0$$

$$\text{given } x = \frac{L}{2}$$

$$mg \frac{L}{2} = P_0AL \ln \left(\frac{L}{L-L/2} \right)$$

$$\frac{4}{3}\pi r^3 dg \frac{1}{2} = P_0\pi r^2 \ln 2$$

$$r = \frac{3}{2} \frac{P_0 \ln 2}{dg}$$

(b) In equilibrium.

$$P\pi r^2 = mg \quad P_0 \left(\frac{L}{L-x} \right) \pi r^2 = \frac{4}{3}\pi r^3 dg$$

$$x = L \left(1 - \frac{1}{2\ln 2} \right).$$

4. **Ans (B,C)**

Using energy conservation

$$\frac{1}{2} \times \frac{5m\ell^2\omega^2}{4} = \frac{3mg\ell}{\sqrt{3}}$$

$$\omega = \sqrt{\frac{8\sqrt{3}g}{5\ell}}$$

$$\text{Magnetic moment } M = \frac{\sqrt{3}}{4} i\ell^2$$

$$\tau = MB = \frac{\sqrt{3}}{\ell} Bi\ell^2$$

$$\text{Now, } \int \tau dt = I\omega$$

$$\frac{\sqrt{3}}{4} B \cdot \ell^2 \int i dt = \frac{5m\ell^2}{4} \omega$$

$$B = \frac{5m\omega}{\sqrt{3}Q}$$

$$B = \frac{5m}{\sqrt{3}Q} \sqrt{\frac{8\sqrt{3}g}{5\ell}}$$

$$B = \frac{m}{Q} \sqrt{\frac{40g}{\sqrt{3}\ell}}$$

5. **Ans (A,C)**

When only s_2 is open

$$\frac{2}{3} = \frac{4}{120} \ell \Rightarrow \ell = 20 \text{ cm}$$

when only s_1 is open

$$2 = \left(\frac{4}{120}\right) \ell \Rightarrow \ell = 60 \text{ cm}$$

when all three are closed

$$2 = \frac{4}{120} \times \ell \Rightarrow \ell = 60 \text{ cm}$$

when only s_1 is closed

6. **Ans (A,B)**

$$(A) \lambda_i = \frac{10u - \left(\frac{u}{2}\right)}{f} = \frac{19u}{2f}$$

$$(B) f_i = \frac{10u - u}{10u - \frac{u}{2}} \cdot f = \frac{18f}{19}$$

$$(C) \lambda_r = \frac{10u + u}{f_r} = \frac{11u \times 19}{18f} = \frac{11 \times 19u}{18f}$$

$$(D) f_r = \frac{18f}{19}$$

PART-1 : PHYSICS

SECTION-I (ii)

7. **Ans (B)**

Consider a point P on screen

$$\text{Optical path length, } [S_1 P]_{\text{liquid}} = [\mu S_1 P]_{\text{air}}$$

$$[S_1 P - t]_{\text{liquid}} + t_{\text{glass}} = \mu_1 [S_2 P - t]_{\text{air}} + [\mu_g t]_{\text{air}}$$

$$= [\mu_1 S_2 P + (\mu_g - \mu_1) t]_{\text{air}}$$

Hence optical path difference at P,

$$\Delta x = [\mu_1 S_2 P + (\mu_g - \mu_1) t] - \mu_1 S_1 P$$

$$= \mu_1 (S_2 P - S_1 P) + (\mu_g - \mu_1) t$$

For a point P at the screen in the absence of liquid

$$S_2 P = S_1 P = \frac{yd}{D}$$

If a liquid is filled

$$[S_2 P - S_1 P]_{\text{liquid}} = \frac{\mu_1 y \cdot d}{D}$$

$$\text{Thus, } \Delta x = \frac{\mu_1 yd}{D} + [\mu_g - \mu_1] t$$

For central maxima

$$\Delta x = 0$$

$$y = - \left[\frac{\mu_g - \mu_1}{\mu_1} \right] \frac{tD}{d} = \frac{(4 - T)tD}{(10 - T)d}$$

at point O, $y = 0$

Thus $T = 4 \text{ sec}$

8. **Ans (A)**

Speed of central maxima

$$v = \left| \frac{dy}{dT} \right| = \frac{6Dt}{(10 - T)^2 d}$$

central maxima is at O at time $T = 4 \text{ sec}$

$$v = 3 \times 10^{-3} \text{ m/sec}$$

9. **Ans (B)**

$$N \sin \theta = \frac{mv_h^2}{r_0}$$

$$N \cos \theta = mg$$

$$\Rightarrow \tan \theta = \frac{v_h^2}{gr_0} = \frac{dz}{dr} = 2 kr_0$$

$$z_0 = kr_0^2$$

$$\Rightarrow v_h = \sqrt{2gz_0}$$

10. **Ans (A)**

Conservation of energy :

$$\frac{1}{2}mv_0^2 + mgz_0 = \frac{1}{2}mv^2 + mgz$$

Conservation of angular momentum about the axis of the bowl :

$$mv_0r_0 = mvr$$

$$z_0 = kr_0^2$$

$$\Rightarrow (z - z_0) \left(z - \frac{v_0^2}{2g} \right) = 0$$

PART-1 : PHYSICS

SECTION-II (i)

2. **Ans (8.00)**

at $v = v_\tau$

Power given = power loss

$$F v_\tau = I^2 \left(1 + \frac{4}{4} \right)$$

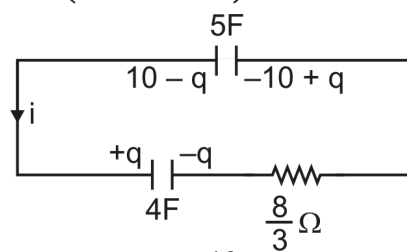
$$I = \frac{B v_\tau (2r)}{\left(1 + \frac{4}{4} \right)}$$

By solving $v_\tau = 64$ m/s
and $I = 8$ A

3. **Ans (1.10 to 1.12)**

$$\frac{10}{9}C = 1.11$$

4. **Ans (4.43 to 4.45)**



$$-\frac{q}{4} - i \times \frac{8}{3} + \frac{10-q}{5} = 0$$

$$\int_0^q \frac{dq}{2 - \frac{9}{20}q} = \int_0^t \frac{3}{8} dt$$

$$q = \frac{40}{9} \left(1 - e^{-\frac{27}{160}t} \right)$$

at steady state

$$q = \frac{40}{9}C$$

the charge on capacitor x at steady state

$$= \frac{1}{4} \times \frac{40}{9} = \frac{10}{9} = 1.11 C$$

$$U_i = \frac{(10)^2}{2 \times 5}$$

$$U_f = \frac{(40/9)^2}{2 \times 4} + \frac{(50/9)^2}{2 \times 5}$$

$$\Delta U = U_i - U_f = \frac{40}{9} J.$$

5. **Ans (1.20 to 1.30)**

$$mg = F_b$$

$$a^3 \times 0.4 \rho g = a^2 h \rho g$$

$$h = \frac{2}{5} a$$

$$m = a^3 \times 0.4 \rho$$

$$F_{\text{net}} = a^2 \times \rho g$$

$$T = 2\pi \sqrt{\frac{a^3 \times 0.4 \rho}{a^2 \rho g}}$$

$$T = 2\pi \sqrt{\frac{2a}{5g}}$$

6. **Ans (0.40)**

$$0.4 \text{ m}$$

Displacement must be less than submergence depth of cube.

PART-1 : PHYSICS

SECTION-II (ii)

7. **Ans (3)**

$$A_s \phi = \frac{d\delta}{d\lambda}$$

$$\phi = \frac{d\delta}{d\lambda} \cdot \frac{d\mu}{d\lambda} = \frac{d\delta}{d\mu} \times \frac{d\mu}{d\lambda}, \delta = (\mu - 1)A_0$$

$$\frac{d\delta}{d\mu} = A_0$$

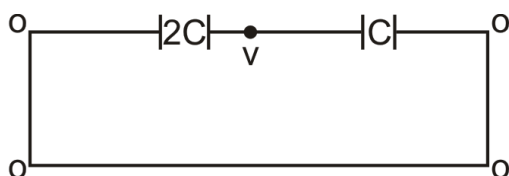
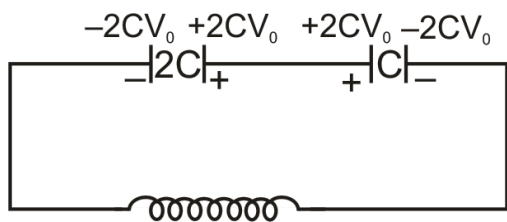
$$\phi = A_0 \frac{d\mu}{d\lambda}$$

$$\mu = A + \frac{B}{\lambda^2}$$

$$\frac{d\mu}{d\lambda} = 0 - \frac{2B}{\lambda^3}$$

8. Ans (8)

$t = 0$



when current is maximum

$$\frac{di}{dt} = 0$$

$$L \frac{di}{dt} = 0$$

i.e. voltage across inductor is zero.

so these two capacitor can be assumed to be connected in parallel.

$$2CV + CV = 4 CV_0$$

$$3CV = 4 CV_0$$

$$V = \frac{4 CV_0}{3}$$

by conservation of energy.

$$\frac{1}{2} \times 2CV_0^2 + \frac{1}{2} \times C.(2V_0)^2 = \frac{1}{2} (2C + C)V^2 + \frac{1}{2} LI^2$$

$$2CV_0^2 + 4CV_0^2 = 3CV^2 + LI^2$$

$$6CV_0^2 - 3CV^2 = LI^2$$

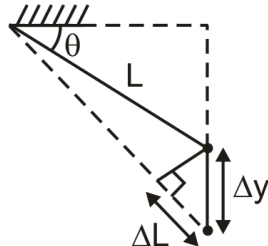
$$6CV_0^2 - 3C \cdot \frac{16V_0^2}{9} = LI^2$$

$$6CV_0^2 - \frac{16CV_0^2}{3} = LI^2$$

$$\frac{2}{3} CV_0^2 = LI^2$$

$$I = \sqrt{\frac{2 CV_0^2}{3 L}} = \sqrt{\frac{2}{3} \times \frac{2 \times 16}{3}} = \frac{2 \times 4}{3} = \frac{8}{3} \text{ A}$$

9. Ans (8)



Length of string changes by $\Delta L = L\alpha_2\Delta T$

the wooden plank descends by

$$\Delta y = \frac{\Delta L}{\sin \theta}$$

$$\sin \theta = 1/2$$

$$\Delta y = 2\Delta L$$

Change in radius of ball $\Delta R = R\alpha_1\Delta T$

$\therefore$ the centre of ball does not move so $\Delta y = \Delta R$

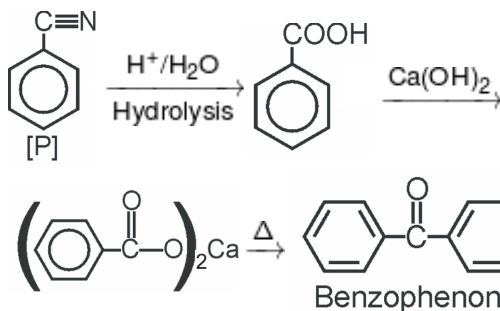
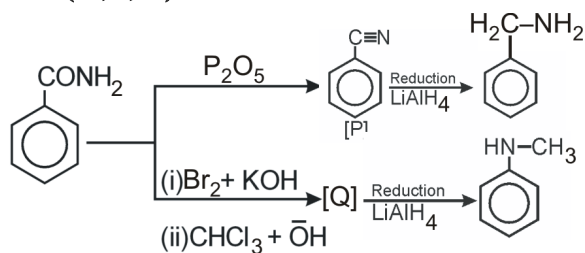
$$2L\alpha_2\Delta T = R\alpha_1\Delta T \Rightarrow 8R\alpha_2 = R\alpha_1$$

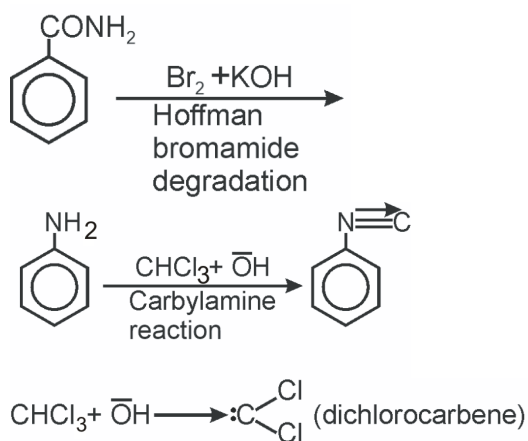
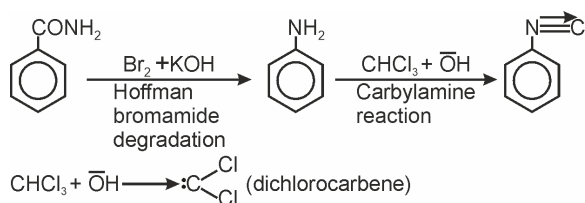
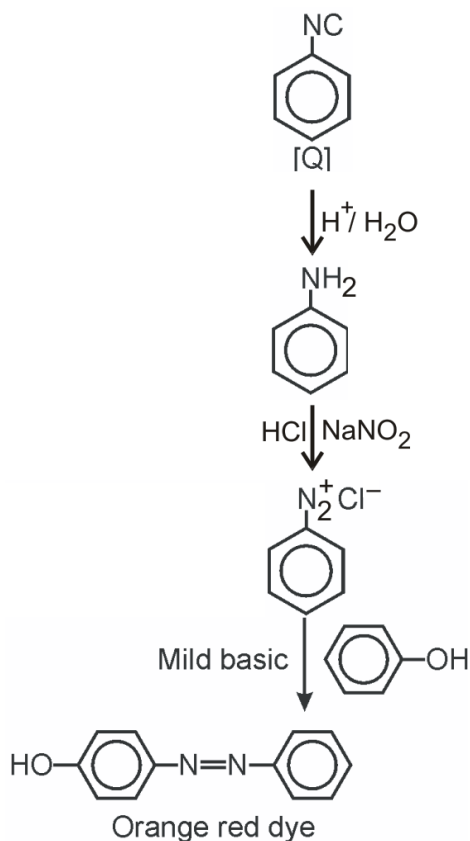
$$\frac{\alpha_2}{\alpha_1} = \frac{1}{8} = 0.125$$

PART-2 : CHEMISTRY

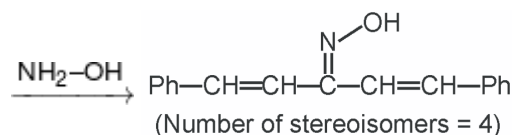
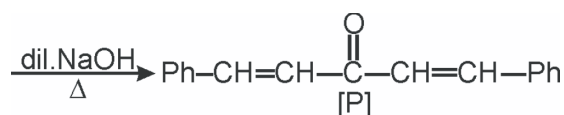
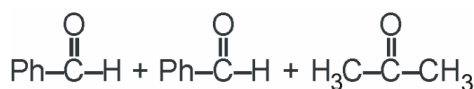
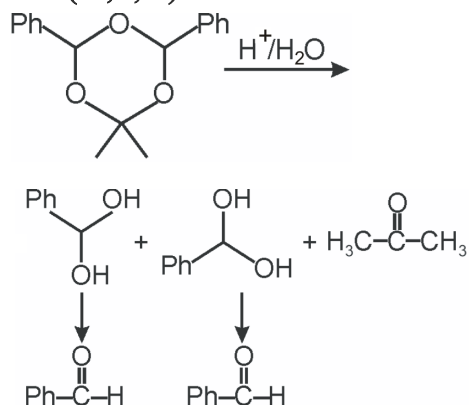
SECTION-I (i)

5. Ans (B,C,D)





6. Ans (A,B,C)



PART-2 : CHEMISTRY

SECTION-I (ii)

8. Ans (C)

$$PV = nRT$$

$$P = \frac{nR}{V} T$$

$$y = mx.$$

$$\text{Slope} = \frac{nR}{V}$$

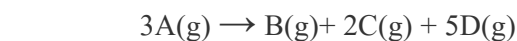
$$\text{For } T_1 - T_2$$

$$\text{Slope} = \tan 55^\circ = \frac{nR}{V}$$

$$1.43 = \frac{nR}{V}$$

$$n = \frac{1.43 \times 0.0821}{0.0821}$$

$$n = 1.43$$



$$\text{at } t = 0 \quad 1 \quad 0 \quad 0 \quad 0$$

$$\text{at } t = t \quad 1-\alpha \quad \frac{\alpha}{3} \quad \frac{2\alpha}{3} \quad \frac{5\alpha}{3}$$

$$\therefore 1 - \alpha + \frac{\alpha}{3} + \frac{2\alpha}{3} + \frac{5\alpha}{3} = 1.43$$

$$\alpha = 0.258$$

PART-2 : CHEMISTRY

SECTION-II (i)

1. **Ans (26.00)**

(i) $M(AB)_2(CD) \rightarrow GI = 4$, Optically Active = 8

(ii) $M(AB)_2cd \rightarrow GI = 6$, Optically Active = 10

(iii) $M a_2b_2c_2 \rightarrow GI = 5$, Optically Active = 2

(iv) $M(AB)_2c_2de \rightarrow GI = 7$, Optically Active = 10

3. **Ans (75.38)**

$$\Delta H_{400} = \Delta H_{300} + \Delta C_p(T_2 - T_1)$$

$$\Delta H_{400} = \Delta H_{300} + \frac{(40 - 50) \times 100}{1000} = 29 \text{ kJ}$$

$$\Delta S_{\text{vap}}(400 \text{ K}) = \frac{29}{400} = 0.0725 \times 1000 = 72.5 \text{ J/K - mol}$$

$$\Delta S_{400} = \Delta S_{300} + \Delta C_p \ln \frac{T_2}{T_1}$$

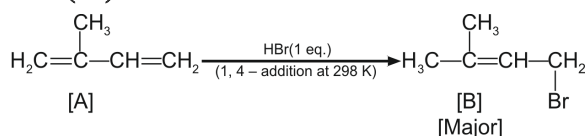
$$72.5 = \Delta S_{300} - 10 \times \ln \left(\frac{4}{3} \right)$$

$$\Delta S_{300} = 75.38 \text{ J/K - mol}$$

PART-2 : CHEMISTRY

SECTION-II (ii)

7. **Ans (5)**



8. **Ans (7)**



9. **Ans (6)**

$$\Delta T_B = i \times K_b \times m$$

$$\frac{103}{200} = i \cdot \frac{(0.57) \times 1000 \times 15}{(342)(85)}$$

$$i = 1.75.$$

$$\alpha = \frac{i - 1}{n - 1} = 0.75$$

$$t_{75\%} = 10 \text{ hours} = 2t_{1/2}$$

$$t_{87.5\%} = 3t_{1/2} = 3 \times 5 = 15 \text{ hours}$$

$$\frac{T}{2.5} = \frac{15}{2.5} = 6$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (B,D)**

Let $P(E_k) = \lambda k$ where λ = proportionality constant

$$\therefore P(E_1) + P(E_2) + \dots + P(E_n) = 1$$

$$\Rightarrow \lambda \cdot 1 + \lambda \cdot 2 + \lambda \cdot 3 + \dots + \lambda \cdot n = 1$$

$$\Rightarrow \frac{\lambda (n(n+1))}{2} = 1 \Rightarrow \lambda = \frac{2}{n(n+1)}$$

$$\text{and } P(A) = P(E_1) P(A/E_1) + P(E_2) P(A/E_2) + \dots + P(E_n) P(A/E_n)$$

$$= \lambda \cdot 1 \cdot \frac{1}{n} + \lambda \cdot 2 \cdot \frac{2}{n} + \dots + \lambda \cdot n \cdot \frac{n}{n}$$

$$= \frac{\lambda}{n} (1^2 + 2^2 + \dots + n^2) =$$

$$\frac{2}{n^2(n+1)} \cdot \frac{n(n+1)(2n+1)}{6}$$

$$= \frac{2n+1}{3n}$$

$$\therefore \lim_{n \rightarrow \infty} P(A) = \frac{2}{3}$$

$$P(E_1/A) = \frac{P(E_1) P(A/E_1)}{P(A)} = \frac{\lambda \cdot 1 \cdot \frac{1}{n}}{\frac{(2n+1)}{3n}}$$

$$= \frac{6}{n(n+1)(2n+1)}$$

2. Ans (A,B,D)

Let $C = (\lambda_1 + 1, 2\lambda_1 + 2, 3\lambda_1 + 3)$ &

$D = (\lambda_2 + 1, 2\lambda_2 + 2, 3\lambda_2 + 3)$

$\therefore CD = \sqrt{14}$

$\Rightarrow (\lambda_1 - \lambda_2)^2 + 4(\lambda_1 - \lambda_2)^2 + 9(\lambda_1 - \lambda_2)^2 = 14$

$\Rightarrow \lambda_1 - \lambda_2 = +1 \text{ or } -1$

$\Rightarrow \lambda_1 = \lambda_2 + 1 \text{ or } \lambda_2 - 1$

Due to symmetrically, both will give same locus

$\therefore$ taking $\lambda_2 = \lambda_1 + 1$

Let centroid of tetrahedron is $G(\alpha, \beta, \gamma)$

$\therefore \alpha = \frac{4 + \lambda_1 + \lambda_2}{4} \Rightarrow 4\alpha = 5 + 2\lambda_1$

$\beta = \frac{8 + 2\lambda_1 + 2\lambda_2}{4} \Rightarrow 4\beta = 10 + 4\lambda_1$

and $\gamma = \frac{6 + 3\lambda_1 + 3\lambda_2}{4} \Rightarrow 4\gamma = 9 + 6\lambda_1$

$\therefore$ Locus of (α, β, γ) is

$\frac{4x - 5}{2} = \frac{4y - 10}{4} = \frac{4z - 9}{6}$

$\Rightarrow \frac{x - \frac{5}{4}}{1} = \frac{y - \frac{5}{2}}{2} = \frac{z - \frac{9}{4}}{3} = \lambda$

$\therefore$ a point having x coordinate as $\left(\frac{3}{2}\right)$ lies on it

then $\lambda = \frac{3}{2} - \frac{5}{4} = \frac{1}{4}$

$\therefore y = \frac{5}{2} + \frac{1}{2} = 3$ & $z = \frac{9}{4} + \frac{3}{4} = 3$

$\therefore$ Equation of locus will be

$\frac{x - \frac{3}{2}}{1} = \frac{y - 3}{2} = \frac{z - 3}{3}$

3. Ans (A,D)

$\therefore \sum_{K=1}^{1008} |z^{2K+1} - z^{2K-1}| = \sum_{K=1}^{1008} |z^{2K} - z^{2K-2}|$
 $= \sqrt{2016}$

$\Rightarrow |z| = 1$ & $\sum_{K=1}^{1008} |z^{2K} - z^{2K-2}| = \sqrt{2016}$

$\Rightarrow |z^2 - 1| \sum_{K=1}^{1008} |z|^{2K-2} = \sqrt{2016}$

$\Rightarrow 1008 |z^2 - 1| = \sqrt{2016}$

$\therefore |z| = 1$

$\therefore$ Let $z = \cos\theta + i \sin\theta$

$\therefore |\cos 2\theta + i \sin 2\theta - 1| = \frac{\sqrt{2016}}{1008}$

$\Rightarrow |-2 \sin^2\theta + i \cdot 2 \sin\theta \cos\theta| = \frac{\sqrt{2}}{\sqrt{1008}}$

$\Rightarrow |\sin\theta| = \frac{1}{\sqrt{2016}}$

$\therefore \operatorname{Re}(z^2) = \cos 2\theta$

$= 1 - 2 \sin^2\theta = 1 - \frac{1}{1008} = \frac{1007}{1008}$

4. Ans (A,B,C)

$\therefore f'(x) = 0$ has $x = a$ as repeated roots.

$\therefore f'(x) = 3(x-a)^2$

$\Rightarrow f(x) = (x-a)^3 + b$

$\therefore g(x) = (x-a)^3 - 3(x-a)^3 + 3(x-a)^2$

$= -2(x-a)^3 + 3(x-a)^2$

$g(a) = 0$

$\therefore g(b) = 2(b-a) - 3 = 0 \Rightarrow 2b = 2a + 3$ and $g'(2) = 0$

$g'(x) = -6(x-a)^2 + 6(x-a) = -6(x-a)(x-a-1)$

$\Rightarrow g'(2) = 0 \Rightarrow 2 - a - 1 = 0 \Rightarrow a = 1$

$\therefore b = \frac{5}{2}$

$\therefore f''(x) = 6(x-a), f''(2) = 6(2-1) = 6$

$\int_0^2 f(x) dx = \int_0^2 \left((x-1)^3 + \frac{5}{2} \right) dx = 5 = 2b$

Clearly, $f(x)$ is always increasing $f(x) = 0$ has only

one real root $\left(\frac{-5}{2}\right)^{\frac{1}{3}} + 1$

$\therefore f(f(x)) = 0 \Rightarrow f(x) = \left(\frac{-5}{2}\right)^{\frac{1}{3}} + 1$ will have only one solution.

5. Ans (A,B,C)

$$f(x) = \lim_{\alpha \rightarrow 0} \frac{e^x (e^\alpha - 1) \cdot \sin \alpha \cdot \ln(1 + \alpha) (2017\alpha^3 + 4)}{4\alpha^3 (2018\alpha^3 + 1)}$$

$$= e^x$$

$$\text{and } g(x) = x^2 + x \int_{-1}^1 g(y) dy + \int_{-1}^1 yg(y) dy$$

$$\text{Let } A = \int_{-1}^1 g(y) dy \text{ and } B = \int_{-1}^1 yg(y) dy$$

$$\therefore g(x) = x^2 + Ax + B$$

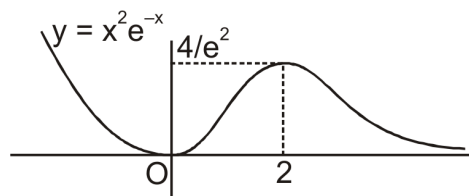
$$\therefore A = \int_{-1}^1 (y^2 + Ay + B) dy = 2 \left(\frac{y^3}{3} + By \right)_0^1$$

$$= \frac{2}{3} + 2B$$

$$\Rightarrow A - 2B = \frac{2}{3}$$

$$\text{and } B = \int_{-1}^1 (y^3 + Ay^2 + By) dy = 2 \left(\frac{Ay^3}{3} \right)_0^1$$

$$= \frac{2A}{3}$$



$$\therefore A - \frac{4A}{3} = \frac{2}{3} \Rightarrow A = -2 \text{ and } B = \frac{-4}{3}$$

$$\therefore g(x) = x^2 - 2x - \frac{4}{3} = (x-1)^2 - \frac{7}{3}$$

$$\therefore f(x) = g(x+1) + \frac{7}{3}$$

$\therefore$ Clearly number of solutions = 1.

6. Ans (A,B,D)

$$\int_0^x t \cdot f(x-t) dt = \int_0^x f(t) dt + \sin x + \cos x - x - 1$$

King

$$\int_0^x (x-t) f(t) dt = \int_0^x f(t) dt + \sin x + \cos x - x - 1$$

$$x \int_0^x f(t) dt - \int_0^x t \cdot f(t) dt$$

$$= \int_0^x f(t) dt + \sin x + \cos x - x - 1$$

Differentiable (Leibniz rule)

$$x f(x) + \int_0^x f(t) dt - x f(x) = f(x) + \cos x - \sin x - 1$$

$$\int_0^x f(t) dt = f(x) + \cos x - \sin x - 1$$

$$\Rightarrow x=0, f(0)=0$$

Again differentiate,

$$f(x) = f'(x) - \sin x - \cos x$$

$$\frac{dy}{dx} - y = \sin x + \cos x$$

Linear differential equation : I.F. = $e^{\int -1 dx} = e^{-x}$

$$y \cdot e^{-x} = \int e^{-x} (\sin x + \cos x) dx + C$$

$$= -e^{-x} \cos x + C$$

$$y = Ce^x - \cos x \Rightarrow C = 1$$

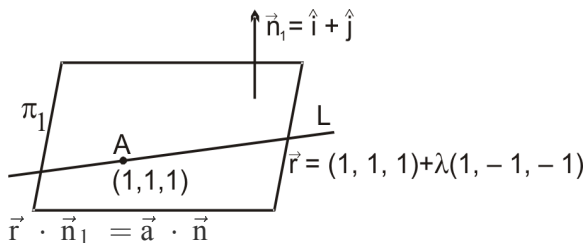
$$y = e^x - \cos x.$$

PART-3 : MATHEMATICS

SECTION-I (ii)

7. **Ans (A)**

$$(\vec{r} - \vec{a}) \cdot \vec{n}_1 = 0$$



$$(x\hat{i} + y\hat{j} + z\hat{k}) \cdot (\hat{i} + \hat{j}) = 2$$

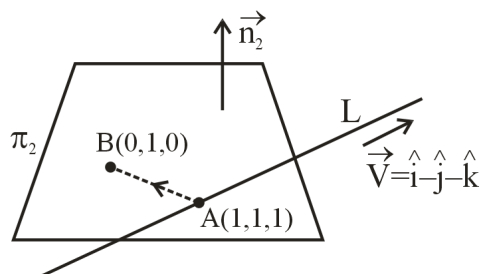
$$x + y = 2 \dots\dots\dots \text{plane } \pi_1$$

$$\text{Now, } \vec{AB} = \hat{i} + \hat{k}; A(1, 1, 1); B(0, 1, 0)$$

$$\text{Now, } \vec{AB} \times \vec{V} = \vec{n}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 0 & 1 \\ 1 & -1 & -1 \end{vmatrix}$$

$$\vec{n}_2 = \hat{i}(0 + 1) - \hat{j}(-1 - 1) + \hat{k}(-1)$$

$$= \vec{n}_2 = \hat{i} + 2\hat{j} - \hat{k}$$



Hence, equation of plane π_2 is $(\vec{r} - \hat{j}) \cdot (\hat{i} + 2\hat{j} - \hat{k}) = 0$ or $x + 2y - z = 2$

Vector along the line of intersection of planes

π_1 and π_2 is $\vec{n}_1 \times \vec{n}_2$ which is also perpendicular to the normal vector of plane π_1

$$\therefore \vec{n}_1 \times \vec{n}_2 = -\hat{i} + \hat{j} + \hat{k}$$

$$\text{So, required vector} = \pm \sqrt{6} \left(\frac{-\hat{i} + \hat{j} + \hat{k}}{\sqrt{3}} \right)$$

$$= \pm \sqrt{2}(-\hat{i} + \hat{j} + \hat{k}) \text{ or } \pm \sqrt{2}(-1, 1, 1)$$

8. **Ans (B)**

If θ is the acute angle between π_1 and π_2 then

$$\Rightarrow \cos \theta = \frac{|\vec{n}_1 \cdot \vec{n}_2|}{|\vec{n}_1| |\vec{n}_2|}$$

$$= \left| \frac{(\hat{i} + \hat{j}) \cdot (\hat{i} + 2\hat{j} - \hat{k})}{\sqrt{2} \cdot \sqrt{6}} \right|$$

$$= \frac{3}{\sqrt{2} \cdot \sqrt{6}} = \frac{\sqrt{3}}{2}$$

$$\Rightarrow \theta = \cos^{-1} \left(\frac{\sqrt{3}}{2} \right) = \frac{\pi}{6}$$

9. **Ans (B)**

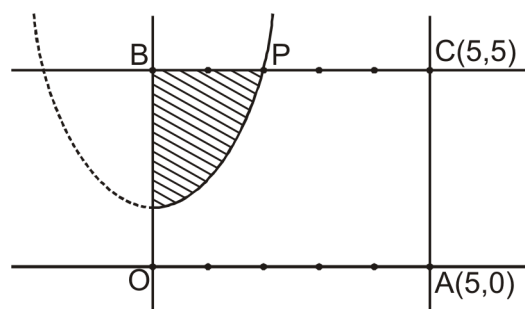
$$\text{Clearly } f(x) = 1 + x^2$$

10. **Ans (A)**

$$y \geq 1 + x^2$$

$$n(S) = 5 \times 5 = 25 \text{ sq. units}$$

$$n(E) = \text{Area of the semi-parabolic region}$$



$$= \int_1^5 x \, dy$$

$$= \int_1^5 \sqrt{y-1} \, dy = \frac{16}{3} \text{ sq. units}$$

$$\text{Required probability} = \frac{n(E)}{n(S)} = \frac{16}{3 \times 25}$$

$$= \frac{1}{3} \left(\frac{f(4) - f(0)}{f(5) - f(0)} \right)]$$

PART-3 : MATHEMATICS

SECTION-II (i)

1. **Ans (5.00)**

Chord subtend right angle at (0, 0) it means it

passes through P(4,0)

$$tx + y = 2t + t^3 \xrightarrow{(4,0)} 4t = 2t + t^3 \Rightarrow t^3 = 2t$$

$$t = 0, \sqrt{2}, -\sqrt{2} \quad A, B, C \equiv (t^2, 2t)$$

$$A(0, 0), B(2, 2\sqrt{2}), C(2, -2\sqrt{2})$$

$$\Delta = \frac{1}{2} \begin{vmatrix} 0 & 0 & 1 \\ 2 & 2\sqrt{2} & 1 \\ 2 & -2\sqrt{2} & 1 \end{vmatrix} = \frac{8\sqrt{2}}{2} = 4\sqrt{2}$$

$$[\Delta] = [4\sqrt{2}] = 5$$

2. **Ans (18.00)**

Chord subtend right angle at (0, 0) it means it

passes through P(4,0)

$$tx + y = 2t + t^3 \xrightarrow{(4,0)} 4t = 2t + t^3 \Rightarrow t^3 = 2t$$

$$t = 0, \sqrt{2}, -\sqrt{2} \quad A, B, C \equiv (t^2, 2t)$$

$$A(0, 0), B(2, 2\sqrt{2}), C(2, -2\sqrt{2})$$

$$t_1 = \sqrt{2}, t_2 = -t_1 - \frac{2}{t_1} \Rightarrow t_2 = -2\sqrt{2}$$

$$B(2, 2\sqrt{2}) \text{ and } B'(8, -4\sqrt{2})$$

$$\ell = BB' = \sqrt{6^2 + (6\sqrt{2})^2} = 6\sqrt{3}$$

$$\ell\sqrt{3} = 18$$

3. **Ans (1.50)**

$$\therefore f(x) = \int_0^{\pi/2} |x - 2 \sin t| \cos t \, dt$$

$$\text{Let } \sin t = z \Rightarrow \cos t \, dt = dz$$

$$\therefore f(x) = \int_0^1 |x - 2z| \, dz = 2 \int_0^1 \left| z - \frac{x}{2} \right| \, dz$$

$$(i) \text{ If } x \leq 0 \text{ then } f(x) = 2 \int_0^1 \left(z - \frac{x}{2} \right) \, dz = 2$$

$$\left(\frac{z^2}{2} - \frac{x}{2}z \right)_0^1 = 1 - x$$

$$(ii) \text{ If } x \geq 2 \text{ then } f(x) = \int_0^1 (x - 2z) \, dz$$

$$= (xz - z^2)_0^1 = x - 1$$

$$(iii) \text{ If } 0 < x < 2, f(x) = 2$$

$$\left(\int_0^{x/2} \left(\frac{x}{2} - z \right) \, dz + \int_{x/2}^1 \left(z - \frac{x}{2} \right) \, dz \right)$$

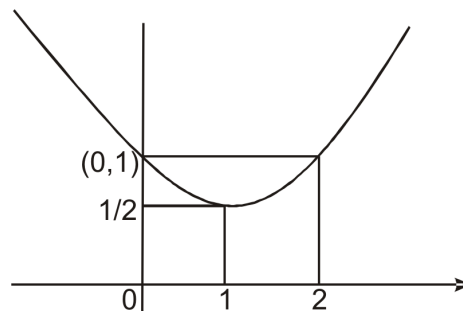
$$= 2 \left(\left(\frac{x}{2}z - \frac{z^2}{2} \right)_0^{x/2} + \left(\frac{z^2}{2} - \frac{x}{2}z \right)_{x/2}^1 \right)$$

$$= \left(\frac{x^2}{2} - \frac{x^2}{4} \right) + \left((1 - x) - \left(\frac{x^2}{4} - \frac{x^2}{2} \right) \right)$$

$$= 1 - x + \frac{x^2}{2}$$

$$\therefore f(x) = \begin{cases} 1 - x, & x \leq 0 \\ \frac{1}{2}(x^2 - 2x + 2), & 0 < x < 2 \\ x - 1, & x \geq 2 \end{cases}$$

Graph of $f(x)$



$\therefore f(|x|) = K$ will have 4 distinct solution

$$\text{for } K \in \left(\frac{1}{2}, 1 \right)$$

4. Ans (2.00)

$$\therefore g(x) = 1 + \int_0^x g(t) dt$$

$$\Rightarrow g'(x) = g(x) \Rightarrow \ln g(x) = x + C$$

$$\therefore g'(0) = 1 \Rightarrow g(0) = 1$$

$$\therefore C = 0$$

$$\therefore g(x) = e^x$$

$$\therefore g(\ln 10) + g'(\ln 10) + g''(\ln 10) = 30$$

$$\therefore \text{Number of solution of } f(x) = 30 \text{ will be } 2$$

5. Ans (300.00)

$$\therefore S = f(1) + 2f(2) + 3f(3)$$

$$\therefore AM \geq GM \Rightarrow \frac{S}{3} \geq (6f(1)f(2)f(3))^{1/3}$$

$$\Rightarrow S^3 \geq 162 f(1)f(2)f(3)$$

$$\text{but given } S^3 = 162 f(1)f(2)f(3)$$

$$\therefore f(1) = 2f(2) = 3f(3)$$

$$\& f(1) + f(2) + f(3) = 1100$$

$$\therefore f(1) = 600, f(2) = 300 \text{ and } f(3) = 200$$

$$\therefore HM = \frac{3}{\frac{1}{f(1)} + \frac{1}{f(2)} + \frac{1}{f(3)}} = \frac{300}{\frac{1}{6} + \frac{1}{3} + \frac{1}{2}} = 300$$

6. Ans (0.45)

$$\text{Probability of white ball} = \frac{2}{11}$$

$$\& \text{probability of not getting white ball} = \frac{9}{11}$$

$$\therefore \text{probability of winning of } B = \frac{9}{11} \times \frac{2}{11} +$$

$$\frac{9}{11} \times \frac{9}{11} \times \frac{9}{11} \times \frac{2}{11} + \dots$$

$$= \frac{\frac{9}{11} \times \frac{2}{11}}{1 - \frac{9}{11} \times \frac{9}{11}} = \frac{18}{40} = \frac{9}{20}$$

PART-3 : MATHEMATICS

SECTION-II (ii)

7. Ans (4)

$$\therefore \text{Area of polygon} = 10 \left(\frac{1}{2} \times 2 \times \cot \frac{\pi}{10} \right)$$

$$= \frac{1}{2} \times 2 \times p_1 + \frac{1}{2} \times 2 \times p_2 + \dots + \frac{1}{2} \times 2 \times p_{10}$$

$$\Rightarrow \therefore p_1 + p_2 + \dots + p_{10} = 10 \cot \frac{\pi}{10}$$

$$\therefore AM \geq HM$$

$$\Rightarrow \frac{p_1 + p_2 + \dots + p_{10}}{10} \geq \frac{10}{\frac{1}{p_1} + \frac{1}{p_2} + \dots + \frac{1}{p_{10}}}$$

$$\Rightarrow \therefore \tan \theta > \theta \quad \forall \quad \theta \in \left(0, \frac{\pi}{2} \right)$$

$$\Rightarrow \tan \frac{\pi}{10} > \frac{\pi}{10} \Rightarrow 10 \tan \frac{\pi}{10} > \pi \quad \dots (2)$$

$$\text{by (1) and (2)}$$

$$\Rightarrow \frac{1}{p_1} + \frac{1}{p_2} + \dots + \frac{1}{p_{10}} > \pi$$

$$\therefore \text{Minimum positive integral value} = 4$$

8. Ans (2)

Straight line $x - y + 1 = 0$ is tangent to the parabola $y^2 = 4x$ at $(1, 2)$

$\therefore (x_0, y_0)$ lying above the line $x - y + 1 = 0$

$\therefore$ required circle touches the parabola and the line at $(1, 2)$

$\therefore$ Family of circle

$$(x-1)^2 + (y-2)^2 + \lambda(x-y+1) = 0$$

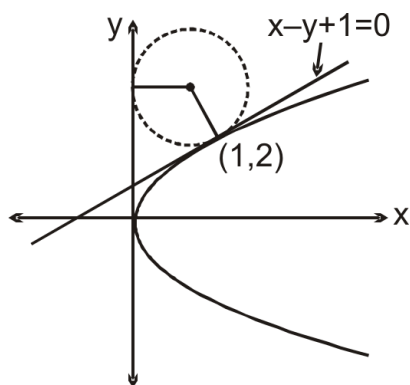
$$x^2 + y^2 + (\lambda-2)x - y(\lambda+4) + 5 + \lambda = 0$$

$\therefore$ circle is also touching the y-axis.

$$\therefore f^2 = c$$

$$\left(\frac{\lambda+4}{2}\right)^2 = 5 + \lambda$$

$$\Rightarrow \lambda^2 + 8\lambda + 16 = 20 + 4\lambda$$



$$\Rightarrow \lambda^2 + 4\lambda + 4 = 8$$

$$\Rightarrow (\lambda + 2)^2 = 8$$

$$\lambda = -2 \pm 2\sqrt{2}$$

$$\therefore x^2 + y^2 + (-4 \pm 2\sqrt{2})x - y(2 \pm 2\sqrt{2}) + 3 \pm 2\sqrt{2} = 0$$

$\therefore (x_0, y_0)$ lying in first quadrant.

$\therefore$ +ve sign is to be taken

$$x^2 + y^2 + (-4 + 2\sqrt{2})x - (2 + 2\sqrt{2})y + 3 + 2\sqrt{2} = 0$$

$$r = x_0 = 2 - \sqrt{2} = \sqrt{2}(\sqrt{2} - 1) = \sqrt{2} \tan \frac{\pi}{8}$$

$$\equiv \sqrt{p} \tan \frac{\pi}{8}$$

$$\therefore p = 2$$

9. Ans (5)

$$x \rightarrow \frac{\pi^-}{2}, t = \tan x \rightarrow \infty$$

$$= \lim_{t \rightarrow \infty} (1+t) \left\{ (1+t) \log \left(1 - \frac{1}{2+t} \right) + 1 \right\}$$

$$= \lim_{t \rightarrow \infty} (1+t) \left[(1+t) \left\{ -\frac{1}{(2+t)} - \frac{1}{2(2+t)^2} - \frac{1}{3(2+t)^3} - \dots \right\} + 1 \right]$$

$$= \lim_{t \rightarrow \infty} (t+1) \left[\left\{ \frac{1+t}{t+2} - \frac{(1+t)}{2(t+2)^2} - \frac{(1+t)}{3(t+2)^3} - \dots \right\} + 1 \right]$$

$$= \lim_{t \rightarrow \infty} (t+1) \left\{ \frac{1}{t+2} - \frac{(1+t)}{2(t+2)^2} - \frac{(t+1)}{3(t+2)^3} - \dots \right\}$$

$$= \lim_{t \rightarrow \infty} \left\{ \frac{t+1}{t+2} - \frac{(t+1)^2}{2(t+2)^2} - \frac{(t+1)^2}{3(t+2)^3} - \dots \right\}$$

$$= 1 - \frac{1}{2} = \frac{1}{2} \equiv \frac{a}{b}$$

$$a = k, b = 2k$$

$$a^2 + b^2 = 5k^2$$

$$a^2 + b^2|_{\min.} = 5$$